



Determination of Thermal Parameters using Thermography and Data Assimilation and its Application to the Convective Heat Transfer Coefficient

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Abstract

This paper presents a method to determine thermal parameters, which can neither be calculated analytically nor measured directly. Specifically, the convective heat transfer coefficient is discussed, which is usually determined using empirical models, namely the Nusselt correlations. To overcome the lack of information about the relation between temperature and parameters, the methodology of data assimilation is applied. Therefore, a dynamic model is combined with measurement data from a thermography experiment avoiding interference with the actual process enabling inline parameter identification. The method is first applied on artificially created simulation data and second on real measurement data. This paper shows that the developed method estimates the heat transfer coefficient in agreement with the well-known Nusselt correlations. Moreover, the present work compares different estimation strategies and gives a recommendation regarding state-parameter, pure parameter or combined estimation, including a detailed analysis of the variation of a smoothing parameter.

Keywords Convective heat transfer coefficient · Data assimilation · Finite element method · Parameter estimation · Thermography

1 Introduction

Production processes, such as the curing of composite parts, heat treatment, coating, etc. and its parameters are often investigated in advance in experiments and simulations. Usually these experiments cannot take all quantities into account, that are affecting the process. Especially in case of non-

linear or non-constant parameters or missing inline process monitoring leads to inefficient production processes with significant safety factors in the process parameters, like temperature and time. Without precise knowledge of these quantities these processes consume much more energy and time than necessary. In the interests of sustainability, these safety factors need to be minimized or even eliminated.

Among the quantities that are subjected to changes during such processes, thermophysical and thermal parameters should be mentioned in particular. These can change for example due to heat treatment or polymer curing, as shown in [1] for magnesium alloys and in [2] for a carbon fiber prepreg, respectively. The parameters describing heat exchange should also be considered, as the processes mentioned previously take place at high temperatures and require heat input, for example by using autoclaves. There is no analytical equation for calculating the convective heat transfer coefficient. With the empirical Nusselt correlations, this parameter can be determined for special geometries and predefined spatial orientations [3]. More general geometric shapes or unsteady flow structures require a separate examination of this parameter, as shown in [4] and [5] respectively.

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However, these empirical models often reach their limits with regard to the processes mentioned initially. Therefore, it is necessary to develop methods which are capable of providing information on the parameters even in these complex non-linear systems. A approach designed for this task is data assimilation (DA), whereby an overview is given in [6]. DA involves the combination of a dynamic model with measurement data in order to enhance the dynamic model in terms of state, parameters or both. The statistical DA methods only work with the currently available measurement data. This enables real-time monitoring of these complex processes, which can take hours and would otherwise require the processing of large amounts of measurement data. The Kalman Filter (KF) and its extensions work according to this principle [7, 8]. The Ensemble Kalman Filter (EnKF) in particular has proven to be a valuable method for recursive filtering for large problems. In [9], the EnKF is used to estimate the internal temperature profile and the thermal conductivity distribution during the curing of carbon fibre-reinforced plastic. Through the EnKF, the thermal conductivity of hyper-eutectic gray cast iron and the heat transfer coefficient between melt and alloy of this material is estimated in [10]. The estimation of the generated curing heat of carbon fiber reinforced polymers by using the EnKF can be found in [11]. Furthermore, the heat transfer coefficients of convection and heating are estimated as constant values therein.

Besides DA, further methods are available for identifying parameters in the context of heat transfer problems, whereby a compact introduction can be found in [12]. Therein occurring Tikhonov regularization method is applied in [13] in combination with fixed-point iteration to determine the regularization parameters as well as the local heat transfer coefficient over the circumference of coiled tubes. For this task, a comparison between the Tikhonov regularization method and the Gaussian filter technique is presented in [14].

In this article, DA in the form of EnKF is utilized for the estimation of thermal parameters as it can handle non-linear systems and provides accurate estimates using only the actual noisy measurement data. The latter in particular is advantageous for the initially mentioned processes, as otherwise large amounts of data have to be stored over a long period of time and the processing of the data requires the process to be finished. Consequently, the searched parameters are also available during the process. Since the focus is on determining thermal parameters, methods for measuring temperatures or derived quantities are to be favored. Furthermore, it is beneficial for the process not to be influenced by the measurements. These requirements are met by thermography, which is a contactless measuring method and has a wide range of applications, e.g. as shown in [15] for condition monitoring of electrical systems, civil structure monitoring or medical applications, etc., to name just a few of the diverse possibilities. Moreover, on the basis of more than

a hundred publications on the signal processing of pulsed thermography for non-destructive evaluation, [16] has presented the latest advances. In the context of thermography and non-destructive evaluation, the work by [17] is particularly noteworthy, which focuses on the determination of material parameters. During all these thermography experiments on a specimen body, the measured thermal radiation emitted is used to make statements about the surface temperature or the heat flow within this body. In order to avoid the negative influence of temperature drift with longer measurement durations, infrared cameras with cooled detectors are in focus here. Moreover, very low measurement noise can be expected with such cameras. However, the EnKF is also capable of dealing with higher measurement noise, as demonstrated in [9]. This method is therefore not limited to a specific detector type, but corrections for temperature drift may be necessary.

As thermography is an imaging process, a separate temperature value is recorded for each measuring point (pixel) at each point in time. The data generated by the chosen measurement method underlines the need for efficient processing, which is ensured by the working principle of the EnKF. In this way, a real-time method for estimating the non-linear course of the investigated parameter is available, whereby different estimation strategies are considered and compared with each other. In order to evaluate this method, the convective heat transfer coefficient is estimated during the cooling process of a flat plate in vertical and horizontal orientation. Thereby, it is assumed that the relation between the searched parameter and the measured temperature is unknown. However, it is possible to identify the non-linear course of the parameter, even if only general reference values are assigned for the initial values of the parameter. By providing parameters in real time, complex processes can be described more precisely, which leads to a reduction in safety factors.

2 Method

A dynamic model used for DA must describe the behavior of the system based on model inputs and model parameters, which may be unknown and therefore must be estimated. Since the task is to estimate thermal parameters, the heat conduction equation

$$\rho c \frac{\partial T(\mathbf{r}, t)}{\partial t} = \nabla \cdot (\Lambda \nabla T(\mathbf{r}, t)) + \dot{g} \quad (1)$$

is selected to determine the spatially and temporally varying temperature $T(\mathbf{r}, t)$ within the domain Ω of the model. The occurring quantities are the time t , the spatial position vector $\mathbf{r}$, the Nabla operator ∇ , the rate of energy production per unit volume $\dot{g}$ and the thermophysical parameters, which

are the density ρ , specific heat capacity c and thermal conductivity λ . The thermal conductivity λ is written here as a second order tensor Λ , enabling the study of anisotropic, orthotropic or isotropic materials. For the derivation and more details on the heat conduction equation, please refer to [18]. The surface of Ω is prescribed with boundary conditions

$$\lambda \frac{\partial T(\mathbf{r}, t)}{\partial \mathbf{n}} = h(T(\mathbf{r}, t)) (T(\mathbf{r}, t) - T_\infty) + \epsilon \sigma (T^4(\mathbf{r}, t) - T_\infty^4) \quad (2)$$

including convection and radiation, which are described by Newton's law of cooling and Stefan-Boltzmann law, respectively. Thereby, $\mathbf{n}$ represents the associated normal vector on the surface of Ω , $h(T)$ the temperature dependent convective heat transfer coefficient, ϵ the emissivity, σ the Stefan-Boltzmann constant and T_∞ the ambient temperature. To solve the transient problem given in Eq. 1, the initial condition

$$T(\mathbf{r}, 0) = T_0(\mathbf{r}) \quad (3)$$

is given by the temperature distribution $T_0(\mathbf{r})$.

The dynamic model is used continuously in the DA procedure, hence it is advisable to adapt it to the problem in focus. Obviously, such simplifications must not lead to any restriction of physically possible phenomena. It is possible to make assumptions using equations which compare different effects. In the case of heat transfer problems, the Biot number

$$Bi = \frac{h \xi}{\lambda} \quad (4)$$

should be mentioned here, in which ξ is half the thickness of the component. This ratio describes the extent to which equalization processes take place with regard to the temperature inside a component in comparison to the temperature difference between the component surface and the environment [19]. With $Bi \gg 1$, large temperature gradients can exist within the component, although the surface temperature is similar to the ambient temperature. In this case, no simplifications are possible, as this would lead to significant deviations from the actual problem. With $Bi \ll 1$, the equalization processes within a component take place much faster than between the component and the environment. As stated in [19], neglecting the temperature gradient in the direction of the component thickness hardly leads to errors, which allows Eq. 1 to be simplified.

2.1 Ensemble Kalman Filter

The estimation of state as well as parameters can be performed in a joint filter (joint estimation) or separately in two filters (dual estimation). According to [20], joint estimation

offers the benefit of representing the dependency between state and parameter in the covariance matrix. The augmented state vector

$$\tilde{\mathbf{x}}^\top = (\mathbf{x}^\top \ \boldsymbol{\theta}^\top) \quad (5)$$

is formed from the state $\mathbf{x}$ and the parameters $\boldsymbol{\theta}$. Depending on the task, $\mathbf{x}$ can describe a location-dependent temperature field or a global temperature value.

The extensions of the KF are intended to handle non-linearities in a system. Let $\mathbf{f}_{k-1}$ describe the state equation and $\mathbf{h}_k$ the measurement equation, a non-linear discrete-time system is then given by

$$\begin{aligned} \tilde{\mathbf{x}}_k &= \mathbf{f}_{k-1}(\tilde{\mathbf{x}}_{k-1}, \mathbf{w}_{k-1}) \\ \mathbf{y}_k &= \mathbf{h}_k(\tilde{\mathbf{x}}_k, \mathbf{v}_k) \end{aligned} \quad (6)$$

where $\tilde{\mathbf{x}}_k$ is the augmented state of the system at time step k , $\mathbf{w}_{k-1}$ the process noise, $\mathbf{y}_k$ the measurement vector and $\mathbf{v}_k$ the measurement noise. For the state, $\mathbf{f}_{k-1}$ is replaced by Eq. 1, but the relation for the parameters is unknown.

In order to process non-linearities, the EnKF uses an ensemble of L members

$$\{\tilde{\mathbf{x}}_k^{(1)}, \tilde{\mathbf{x}}_k^{(2)}, \dots, \tilde{\mathbf{x}}_k^{(l)}, \dots, \tilde{\mathbf{x}}_k^{(L)}\} \quad (7)$$

in which, $\tilde{\mathbf{x}}_k$ of member l is denoted as $\tilde{\mathbf{x}}_k^{(l)}$. The EnKF proceeds forward in time from time step $k-1$ to k by the predictor-corrector method. In the predictor step, the dynamic model is used for each member to generate a forecast value $\tilde{\mathbf{x}}_k^{f(l)}$. The mean value of the ensemble

$$\bar{\mathbf{x}}_k^f = \frac{1}{L} \sum_{l=1}^L \tilde{\mathbf{x}}_k^{f(l)} \quad (8)$$

represents the best estimate at the current time step k . The different types of KF do not assume that the state is perfectly known. The uncertainty regarding the true value is formulated by means of a covariance matrix. In the case of the EnKF, the sample variance of the members is used as process error covariance matrix

$$\mathbf{P}_k^f = \frac{1}{L-1} \sum_{l=1}^L (\tilde{\mathbf{x}}_k^{f(l)} - \bar{\mathbf{x}}_k^f) (\tilde{\mathbf{x}}_k^{f(l)} - \bar{\mathbf{x}}_k^f)^\top \quad (9)$$

In the corrector step, the measurement data is processed to improve $\tilde{\mathbf{x}}_k^{f(l)}$ of the L members. The extent to which the adjustment takes place depends on the Kalman Gain matrix

$$\mathbf{K}_k = \mathbf{P}_k^f \tilde{\mathbf{H}}_k^\top (\tilde{\mathbf{H}}_k \mathbf{P}_k^f \tilde{\mathbf{H}}_k^\top + \mathbf{R}_k)^{-1} \quad (10)$$

where $\mathbf{R}_k$ is the measurement error covariance matrix and $\tilde{\mathbf{H}}_k$ represents the extension of the measurement matrix $\mathbf{H}_k$,

which establishes the correct assignment of state and measurement. As proposed among others in [9, 11], the extended measurement matrix

$$\tilde{\mathbf{H}}_k = (\mathbf{H}_k \mathbf{0}) \quad (11)$$

is composed of $\mathbf{H}_k$ itself and the zero matrix $\mathbf{0}$ in order to express the lack of knowledge about the relationship between the measured state and the estimated parameter. By projecting the innovation onto $\mathbf{K}_k$ the subsequent addition with $\tilde{\mathbf{x}}_k^{f(l)}$ leads to the analysis of $\tilde{\mathbf{x}}_k$

$$\tilde{\mathbf{x}}_k^{a(l)} = \tilde{\mathbf{x}}_k^{f(l)} + \mathbf{K}_k \left(\mathbf{y}_k^{(l)} - \tilde{\mathbf{H}}_k \tilde{\mathbf{x}}_k^{f(l)} \right) \quad (12)$$

whereby the mean value of $\tilde{\mathbf{x}}_k^{a(l)}$ would be the best estimated value of $\tilde{\mathbf{x}}_k$ at time step k . In this paper, the stochastic approach of the EnKF is chosen, which works with perturbed measurement data $\mathbf{y}_k^{(l)}$. This means that a random number $\mathbf{r}_k^{(l)} \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_k)$ is added to $\mathbf{y}_k$. Without perturbation, all members are associated with the same measurement, resulting in an updated ensemble with a too low variance [21].

The parameters θ_k do not receive direct guidance from $\mathbf{y}_k$. Therefore, a smoothing parameter γ ($0 < \gamma < 1$) is introduced, which is used as follows:

$$\theta_k^f = \gamma \theta_{k-1}^f + (1 - \gamma) \theta_{k-1}^a \quad (13)$$

According to [22], $\gamma = 0.8$ is a suitable choice for many experiments. In [11], it is shown that the consideration of the process noise can enhance the results of the estimated parameter.

2.2 Implementation of the Method

Figure 1 shows the workflow of the proposed method to estimate state as well as parameters. The program MATLAB controls the process and ensures that the required quantities are available at the right time step k . The dynamic model is solved with the finite element method in COMSOL Multiphysics. Data exchange between the two programs takes place via the LiveLink interface.

The members of the EnKF are not handled in a programming loop, but are built in the geometry of COMSOL. This leads to lower time durations for computing the solution for one time step.

The predictor-corrector method underlying the EnKF requires initial values $\tilde{\mathbf{x}}_0^{a(l)}$ for state and parameters. The selection can be based on the measurement data [23]. A normal distribution of $\tilde{\mathbf{x}}_0^{a(l)} \sim \mathcal{N}(\mu, \sigma^2)$ is assumed, whereby μ is mean value, σ the standard deviation and σ^2 the variance. Once $\tilde{\mathbf{x}}_0^{a(l)}$ is specified and the individual members are accessible in COMSOL, each time step k is processed in the same pattern. First, COMSOL receives $\tilde{\mathbf{x}}_{k-1}^{a(l)}$ from MATLAB via the LiveLink interface using $\mathbf{T}_k^{f(l)}$ for the initial conditions of each member and $\theta_{k-1}^{a(l)}$ to update the param-

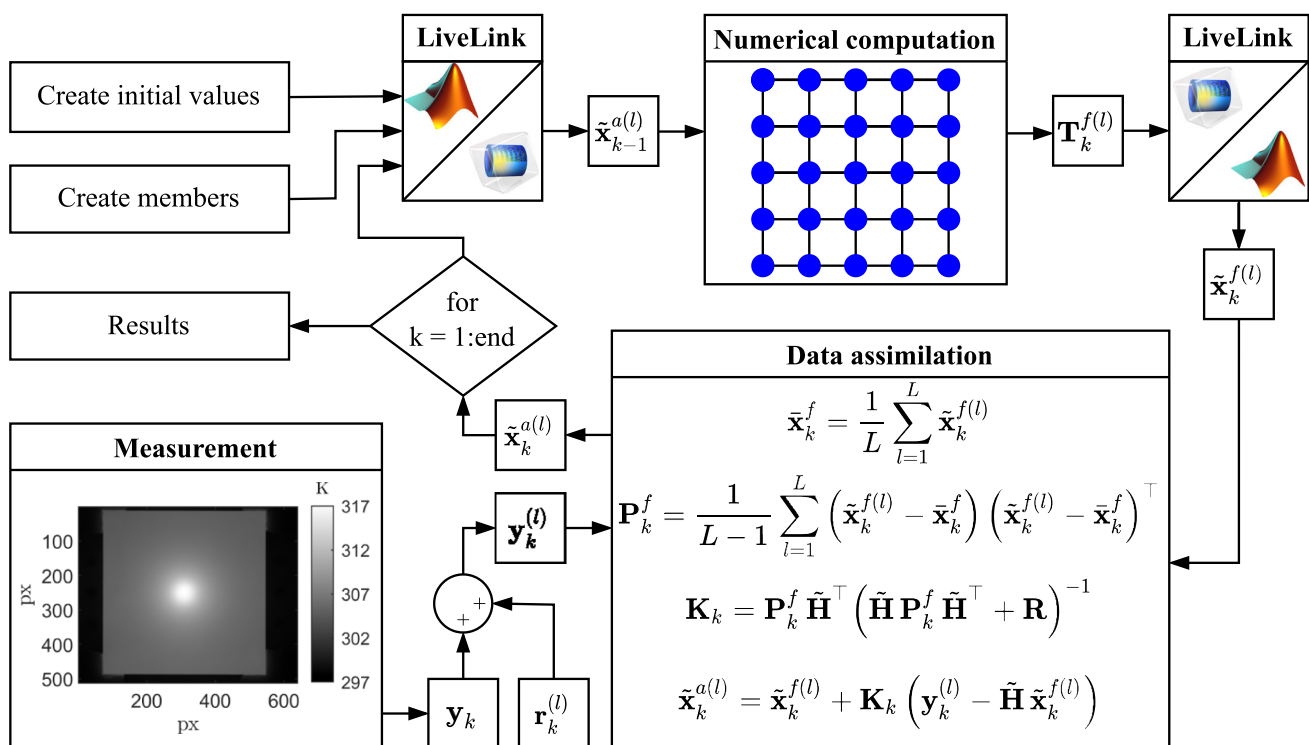


Fig. 1 Workflow of the method through the use of MATLAB and COMSOL Multiphysics

ters. The duration of the numerical computation is equal to the sampling interval of the measurement data. At the end of the numerical computation, the temperature data $\mathbf{T}_k^{f(l)}$ is returned to MATLAB, either as a location-dependent temperature field or as a global scalar quantity. For the next time step k , Eq. 1 only generates forecast values for the state. Although the parameters to be estimated are kept constant during the numerical computation, non-linear courses can be processed by the improvements in the corrector step. The next step is to execute the EnKF procedure, whereby $\mathbf{y}_k^{(l)}$ has to be available. The loop begins again until the final time step is reached.

The workflow presented above is not only designed to perform a combined state-parameter-estimation (SPE), but also a pure parameter estimation (PPE). For the latter, only $\theta_k^{a(l)}$ is calculated by the DA and $\mathbf{T}_{k-1}^{a(l)}$ in COMSOL is equal to the previous calculation result $\mathbf{T}_k^{f(l)}$. SPE provides the most accurate estimate for the state, but this may not give the parameter sufficient error to adequately improve its estimate. On the other hand, PPE can lead to overshoots in the estimation, especially with relatively unknown initial values. Ideally, the strengths of both variants can be combined in order to optimally estimate the searched parameter. Therefore, another implementation is investigated, which performs a SPE whenever θ changes significantly, otherwise only a PPE is performed. This is then referred to as optimal parameter estimation (OPE)

$$\text{OPE} = \begin{cases} \text{PPE} & \text{for } \left| \frac{\theta_{k-1} - \theta_k}{\theta_{k-1}} \right| < \delta \\ \text{SPE} & \text{for } \left| \frac{\theta_{k-1} - \theta_k}{\theta_{k-1}} \right| \geq \delta \end{cases} \quad (14)$$

whereby a relative deviation from the previous time step is formed as the true value of θ is unknown compared to a limit value δ .

3 Estimation of the Convective Heat Transfer Coefficient

The proposed method can be applied to any thermal parameter estimation problems that can be described by a dynamic model. The goal in this section is to use the proposed DA method to estimate the convective heat transfer coefficient h during the cooling of an aluminum plate. The plate is heated approx. 20 K above the ambient temperature T_∞ before the cooling is observed. Since the convective heat transfer coefficient is temperature dependent (i.e. $h(T)$, see Appendix A) it is expected that the estimation method yields non-constant estimates over time. Therefore, this example demonstrates the ability of the method to deal with complex real-world

problems. First, a simulation study with artificial measurement data is carried out to evaluate the method. Additionally, statements about the preferred estimation strategy should be possible. Afterwards, the method and the knowledge gained from the simulation study is applied on real thermography measurement data.

3.1 Experimental Setup and Parameters

The test specimen is a plate (100 x 100 x 1 mm) of the material aluminum AW6061-T6. Table 1 lists the necessary thermophysical parameters, which are the thermal conductivity λ , specific heat capacity c and density ρ . Since different Nusselt correlations are defined for a vertical plate (VP) and horizontal plate (HP), this is also considered in the suggested measurement setups shown in Fig. 2. The contact area between the support and the specimen is kept as small as possible so that the temperature equalization occurs primarily by convection and radiation. Cut-outs in the styrofoam facilitate natural convection which requires undisturbed air flow. Before the workflow in Fig. 1 is triggered, the test specimen is brought out of thermal equilibrium with its environment by heating it up with a heat gun. As the heat input occurs before and not during the DA, no heat source is considered in Eq. 1, i.e., $\dot{g} = 0$.

As already mentioned, it is advisable to adapt the computation dimension of the dynamic model to the task at hand. Since natural convection in air is present, the according range in Table 2 is used for Eq. 4 to compute the Biot number of the test specimen. In the thickness direction of the specimen, the range for h results in a Biot number from $Bi \approx 6 \cdot 10^{-6}$ to $Bi \approx 8 \cdot 10^{-5}$. Consequently, the test specimen is thermally thin ($Bi \ll 1$), enabling a 2D model to be used in which the gradient in the direction of the thickness is neglected. COMSOL supports model reductions from 3D to a lower dimensionality through the out-of-plane heat fluxes [25]. These are used to consider the boundary conditions from Eq. 2. The temperature field of the test specimen is discretised with a mapped mesh. Ten quadrilateral elements are used in each of the two main directions.

The state $\mathbf{x}$ describes the temperature of the test specimen. This can be a location-dependent temperature field $T = T(\mathbf{r}, t)$ or a global temperature value $T = T(t)$. Since the material under investigation has a high λ , equalization processes take place much faster than the cooling process of the plate. This property is utilized to achieve a homogeneous temperature profile through a delay after the heating process

Table 1 Thermophysical parameters AW6061-T6 [24]

T [T] = K	λ [λ] = W/(m K)	c [c] = J/(kg K)	ρ [ρ] = kg/m ³
273.15	162	917	2703

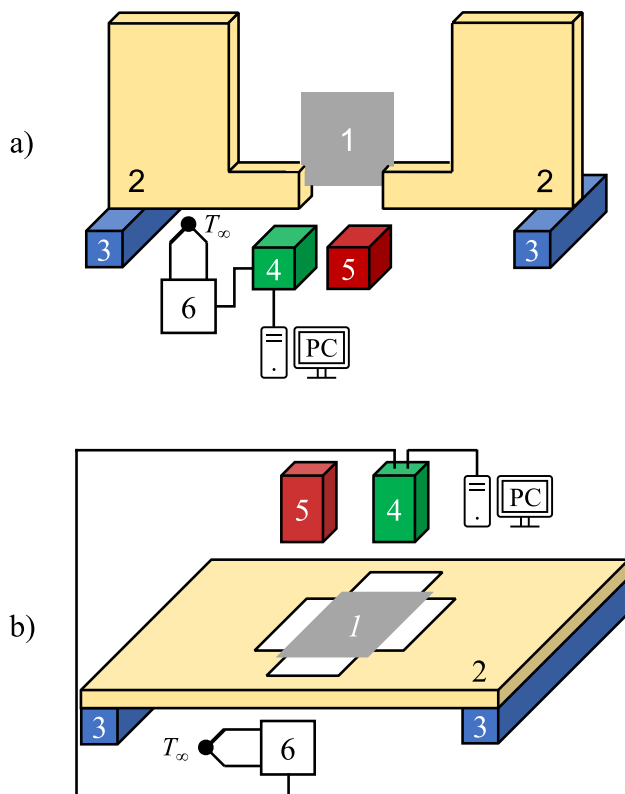


Fig. 2 Suggested measurement setups a) vertical plate and b) horizontal plate: test specimen (1), test specimen support made of styrofoam (2), support structure (3), infrared camera (4), heat gun as heat source (5) and thermocouple (6)

before the DA is triggered. For the simulation study, the initial values are chosen similar to the real measurement data, assuming that the test specimen has a constant temperature of 320 K at this moment.

For this reason, a global temperature value is used for the state $\mathbf{x}$. The parameters θ consist only of $h = h(T)$. For the state, μ and σ are determined from the first frame of the measurement. However, σ is chosen larger than the determined value to evaluate whether convergence to the correct value is possible despite a poorly chosen initial value. Table 2 offers guidance for the choice of $\mathbf{h}_0^{a(l)}$. The mean of the interval is set equal to μ and one half of the interval is selected as the 3σ range, which means that almost all initial values lie in the interval. The specific values describing $\tilde{\mathbf{x}}_0^{a(l)}$ are found in Table 3. In summary, each member is given a spatial equalized temperature profile T_0 for the initial condition in Eq. 3,

Table 2 Typical values for different heat transfer situations [3]

Type and present fluid	h , [h] = W/(m ² K)
Natural convection in gases	2 – 25
Natural convection in liquids	10 – 1000
Forced convection in gases	25 – 250
Forced convection in liquids	50 – 20000

Table 3 Initial values for DA of a VP and HP within the simulation study (Sim) and with real measurement data (Meas)

Data	Setup	Quantity	μ	σ	Unit
Sim	all	$T_0^{a(l)}$	320	0.5	K
Meas	VP	$T_0^{a(l)}$	319	0.5	K
Meas	HP	$T_0^{a(l)}$	320	0.5	K
all	all	$\mathbf{h}_0^{a(l)}$	13.5	3.8	W/(m ² K)

whereby the values for T_0 as well as $h = h(0)$ of each member result from $\mathcal{N}(\mu, \sigma^2)$.

The ensemble of the EnKF consists of 20 members, as trials have shown no improvement with $L > 20$. The measurement sampling rate for both the simulation study and the real measurement data is set to $f = 1$ Hz. In addition,

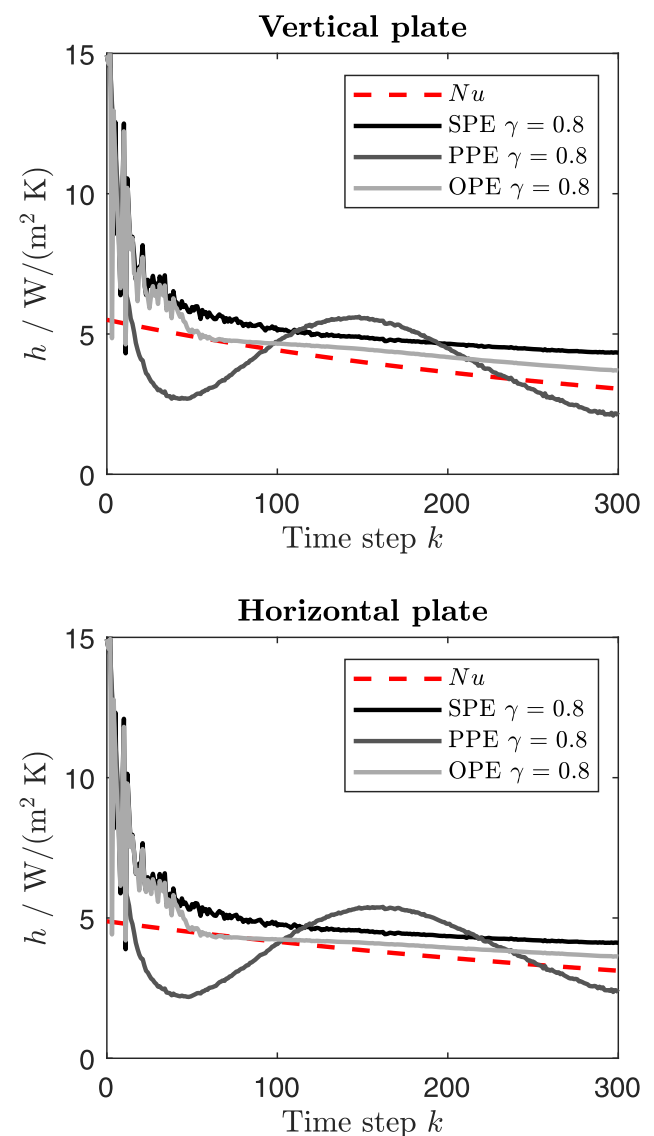


Fig. 3 Estimation values for h within the simulation study - Comparison of SPE, PPE to OPE

the results of the simulation study for the half sampling rate are provided in Appendix B. The limit value in Eq. 14 is $\delta = 0.05$. Both no noise and process noise are studied, whereby the latter is represented as an additive term for $\mathbf{h}_k^f$. The value is taken from a normal distribution, where σ^2 is assumed to be equal to one quarter of σ^2 of $\mathbf{v}_k$.

3.2 Simulation Study

For the simulation study, a numerical computation of the same plate is executed as a 3D shell to generate the artificial measurement data. The initial condition given in Eq. 3 is prescribed as constant temperature field, whereby $T_0(\mathbf{r})$ is the mean value μ of $\mathbf{T}_0^{a(l)}$ (Data Sim) from Table 3. For the boundary conditions in Eq. 2, a constant emissivity $\epsilon = 0.9$

is defined and the convective heat transfer coefficient $h(T)$ is calculated in COMSOL according to the Nusselt correlations in Appendix A. In order to obtain a more realistic behavior, this artificial measurement data is additively perturbed with white noise with a standard deviation $\sigma = 0.05$ K before passing it to the DA.

The estimation results of h within the simulation study are shown for both measurement setups in Figs. 3, 4, and 5. Note that the large deviation between the estimated and target value for h in the first time steps is due to the chosen initial values $\mathbf{h}_0^{a(l)}$, as these were set by the relevant interval from Table 2 and not by the Nusselt correlations.

The different estimation strategies, i.e., SPE, PPE and OPE, are compared with each other in Fig. 3. In both measurement setups, the overshoot is clearly visible when PPE

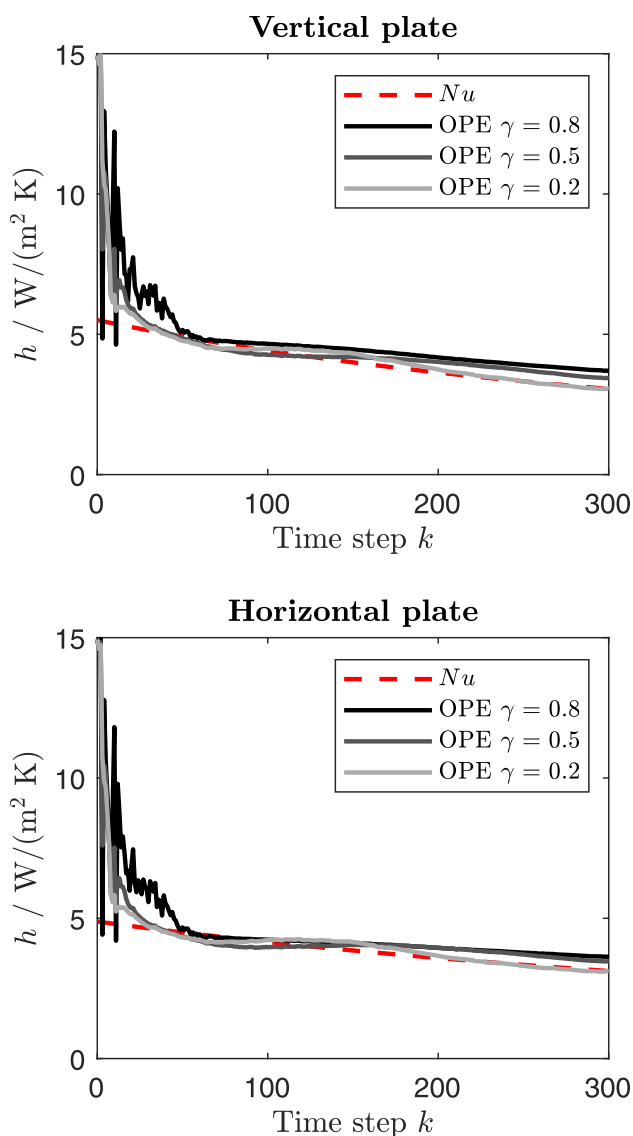


Fig. 4 Estimation values for h within the simulation study - Variation of the smoothing parameter γ with OPE

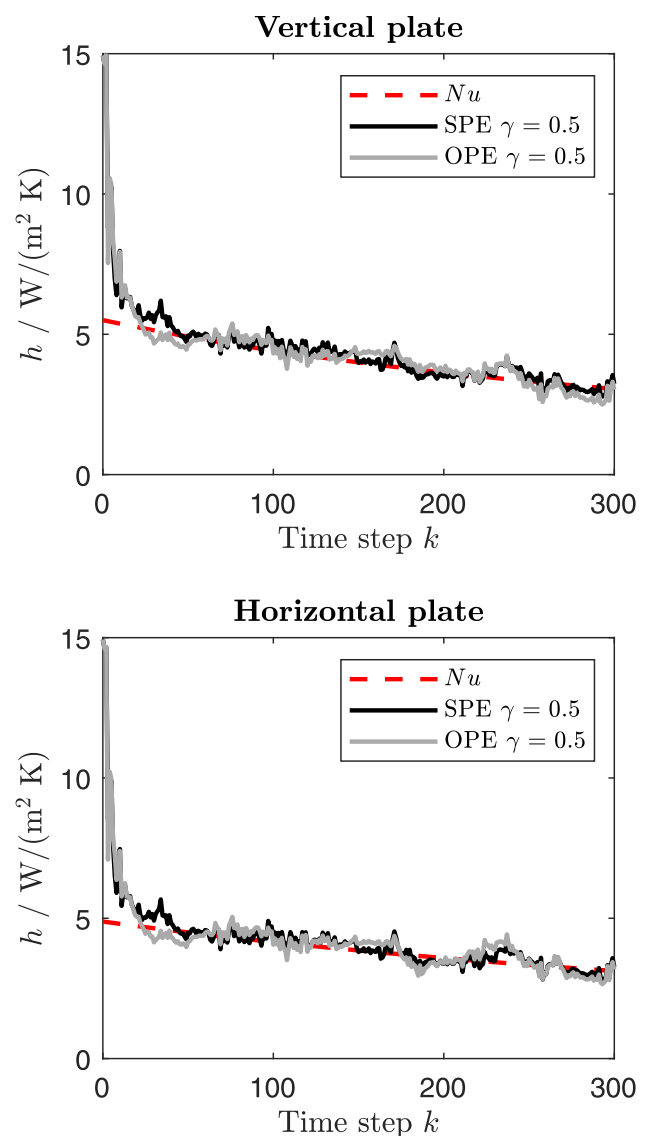


Fig. 5 Estimation values for h within the simulation study - Comparison of SPE and OPE with process noise consideration

Table 4 Time mean error without process noise consideration

Setup	Type	$\epsilon_T (\gamma = 0.8)$ [ϵ_T] = %	$\epsilon_T (\gamma = 0.5)$ [ϵ_T] = %	$\epsilon_T (\gamma = 0.2)$ [ϵ_T] = %	$\epsilon_h (\gamma = 0.8)$ [ϵ_h] = %	$\epsilon_h (\gamma = 0.5)$ [ϵ_h] = %	$\epsilon_h (\gamma = 0.2)$ [ϵ_h] = %
VP	SPE	0.027	0.055	0.083	26.403	26.082	29.078
VP	OPE	0.074	0.028	0.019	12.439	7.627	3.144
HP	SPE	0.022	0.042	0.062	21.048	19.497	21.300
HP	OPE	0.050	0.021	0.016	8.320	7.680	3.110

is used. The reason for this behavior is due to the formulation of convection in the boundary conditions. An incorrect value of h leads to a too small or large heat flux, which the test specimen emits during the cooling process. To eliminate this deficit, the parameter is adapted toward the true value. Since an error in the temperature is constantly caused during this time, h changes until the true value is reached but also the error is eliminated. Now the error is corrected, but the parameter is not at its true value, which causes a change in the opposite direction. Thus, h gradually oscillates to its true value. A clearly different behavior can be found with SPE. By additionally estimating the state, the error in the temperatures is corrected on the spot. This prevents overshooting, but the estimated value only approaches its true value to a limited extent. The better results when using OPE compared to SPE indicates that SPE decreases the error in state estimation sufficiently, whereby the error in h cannot be decreased properly.

Furthermore, the question of whether the change in γ has positive effects is investigated in Fig. 4. Since no improvements can be observed with SPE, only the results of OPE are presented. Lowering γ can lead to an improvement, but the tendency to oscillate increases. It also reduces the number of necessary time steps k until h has reached its true value.

Until now, the process noise of h was ignored. Figure 5 shows the results when the process noise is taken into account as indicated in the method section.

Appendix A gives the relevant Nusselt correlations Nu for the VP and HP cases, from which the theoretical value of h is calculated using the surface temperature at each time step k . The Nusselt correlation for the VP does not distinguish between front and back. In contrast, the upper and lower surfaces of a HP receive different Nusselt correlations. Since the 2D model in COMSOL sums up the out-of-plane heat fluxes, the reference value of the HP is determined as the mean of the two Nusselt correlations. The calculated h is

used as a target value in the simulation study and is plotted as red dashed line in Figs. 3, 4, and 5. It can be seen from the graphs that the estimation using OPE follows the Nusselt correlations the best.

For further evaluation, the time mean error

$$\epsilon_\phi = \frac{1}{N - n} \sum_{k=LB}^{UB} \left| \frac{\phi_k^{true} - \phi_k^a}{\phi_k^{true}} \right| \quad (15)$$

of the quantity ϕ between an lower bound (LB) and an upper bound (UB) is introduced. Here, ϕ_k^a is the estimated value and ϕ_k^{true} the true value at the time step k . In Tables 4 and 5, the evaluation of ϵ_T and ϵ_h for the range [LB; UB] = [60; 300] is presented, whereby the process noise is only considered in the latter.

3.3 Application to Real Data

Figure 6 shows the experimental setup of the thermography experiment using the vertical plate as an example. Essentially, this contains the components mentioned in Fig. 2. The test specimen is placed in a half-closed box to isolate it from the rest of the room. Consequently, in the surroundings of the specimen mainly air flows caused by natural convection due to the higher surface temperature of the test specimen occur. During the measurement, no further heat sources were present within the measuring area. The measurement is carried out with the infrared camera FLIR X6900sc, whereby temperature fluctuations with a standard deviation $\sigma_{IR} = 0.025$ K (NETD) occur [26]. The range [283.15; 363.15] K is selected for the temperature calibration, yielding an integration time of 267.658 μ s. The thermocouple TXDIN1600T is only used to monitor T_∞ .

Thermography experiments on materials with low ϵ can lead to problems with regard to heat input as well as detection of the emitted radiation [27]. Under these conditions, the

Table 5 Time mean error with process noise consideration

Setup	Type	$\epsilon_T (\gamma = 0.8)$ [ϵ_T] = %	$\epsilon_T (\gamma = 0.5)$ [ϵ_T] = %	$\epsilon_T (\gamma = 0.2)$ [ϵ_T] = %	$\epsilon_h (\gamma = 0.8)$ [ϵ_h] = %	$\epsilon_h (\gamma = 0.5)$ [ϵ_h] = %	$\epsilon_h (\gamma = 0.2)$ [ϵ_h] = %
VP	SPE	0.005	0.005	0.005	8.090	5.490	6.081
VP	OPE	0.006	0.006	0.009	8.105	6.862	10.923
HP	SPE	0.005	0.005	0.005	7.034	5.281	6.045
HP	OPE	0.005	0.007	0.008	7.489	7.462	10.774

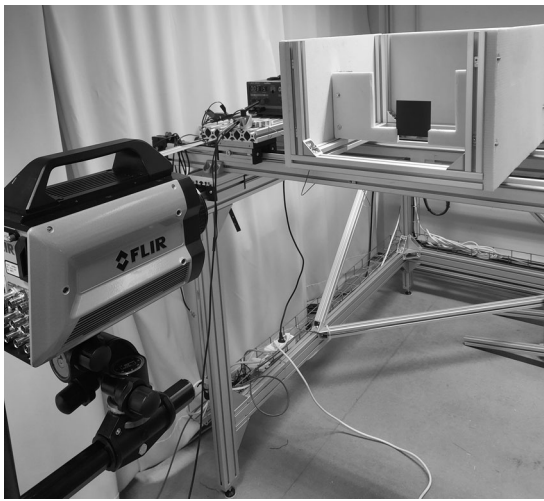


Fig. 6 Measurement setup for the vertical plate

measurement of the actual temperature of the test specimen surface is impeded, as the detected radiation is composed of the radiation emitted by the test specimen itself and the reflected radiation from the environment at the test specimen surface. The composition is equivalent to the relation

$$\alpha + R = 1 \quad (16)$$

in which α is the absorption coefficient and R the reflection coefficient. For this reason, the test specimen is blackened on both surfaces with GRAPHIT 33, a thermoplastic binder with electrically conductive graphite powder. This leads to

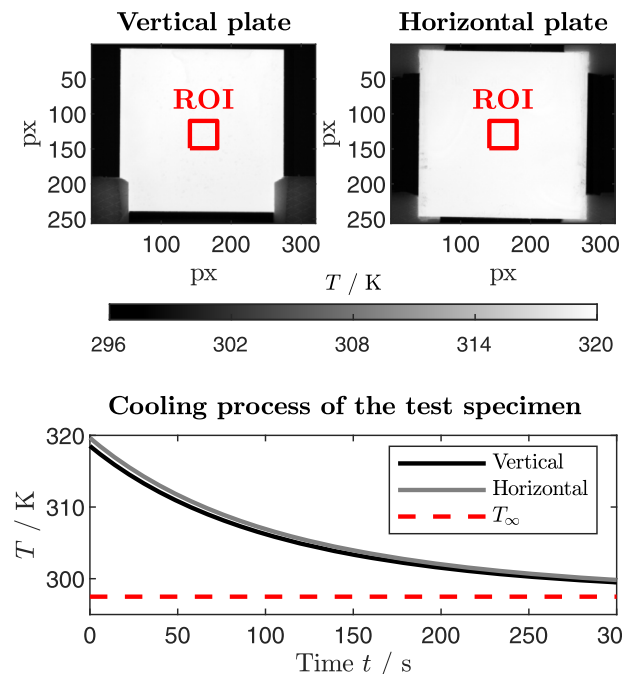


Fig. 7 Workflow for extracting the test specimen temperature T during the cooling process

$\alpha = (91.5 \pm 0.3) \%$ [28]. Consequently, Eq. 16 is dominated by absorption and effects relating to reflection are significantly prevented as the test specimen is positioned as shown in Fig. 6. In order to ensure that these measures are sufficient, the recorded measurement data is verified by subtracting the first frame from all others. By comparing them with the original frames, no unwanted dynamic reflections from the environment are noticed.

Truthfully, the coating with GRAPHIT 33 leads to a multi-layered system. Whether it is permissible to neglect the coating in the dynamic model is to be clarified with the diffusion time

$$\tau_d = \frac{\rho c L_{char}^2}{\lambda} \quad (17)$$

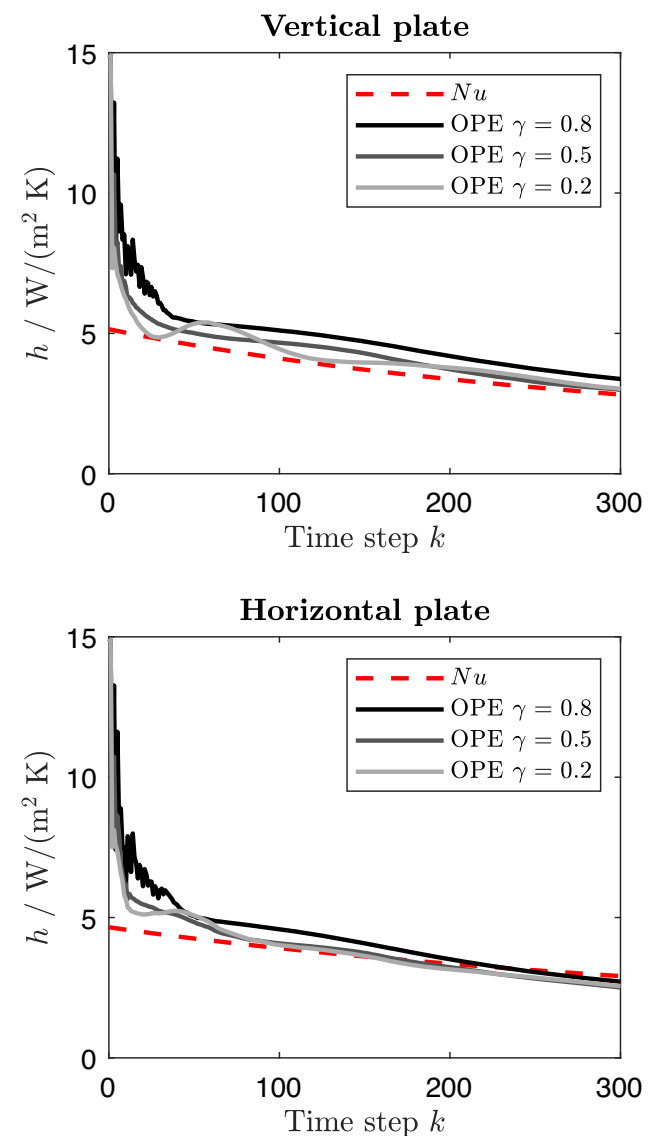


Fig. 8 Estimation values for h with real thermography measurement data - Variation of the smoothing parameter γ with OPE without process noise consideration

in which L_{char} is the characteristic length, i.e., the thickness of the test specimen and graphite layer. The thermophysical parameters of the coating can be found in [29]. The comparison of the diffusion times, $\tau_d = 1.5 \cdot 10^{-2}$ s for the test specimen and $\tau_d = 1.2 \cdot 10^{-4}$ s for the graphite coating, shows a difference of two powers of ten and allows just the test specimen to be modeled in the dynamic model.

Figure 7 shows the first frames of the measurements for the VP and HP as well as the cooling curves extracted from the data. Therefore, the middle pixels within the region of interest (ROI) are averaged. As requested in Section 3.1, the measurement was carried out with a sampling rate $f = 1$ Hz.

The estimation results of h over time with the real thermography measurement data are shown for both measurement setups in Figs. 8 and 9. Figure 8 shows the impact when γ is reduced in OPE, whereby a similar behavior to the simu-

lation study in Fig. 4 is observed. The influence of the system noise is presented in Fig. 9, whereby also a similar behavior compared to the simulation study in Fig. 5 is obtained.

Identical to the simulation study, the Nusselt correlations were used to plot a theoretical h in Figs. 8 and 9 (red dashed line). For this the measured surface temperature at each time step k is used. Since the actual values of the parameters required for the calculation are not known, the theoretical value for h can serve only as a qualitative reference in the graphs. However, it can be seen that the overall trends show a similar behavior.

Besides the measurement data shown here, further series of measurements were recorded, which are in a comparable temperature range. However, different initial temperature values are to be expected for the DA of the individual series. Consequently, at the same time step k , different estimation results for h are obtained for each series. Therefore, the following discussion is based on $h(T)$, as this relation persists through different measurement series. Since all series do not contain the same temperature values, the temperature range [302; 312] K is divided into intervals with a width of 1 K. In each interval, the standard deviation σ_i is determined. These are then combined following [30] to a pooled standard deviation

$$\sigma_p = \sqrt{\frac{\sum_{i=1}^N (n_i - 1) \sigma_i^2}{\sum_{i=1}^N (n_i - 1)}} \quad (18)$$

in which n_i is the number of values within range i and M is the number of ranges. For both measurement setups, Table 6 lists σ_p for the variation of γ in OPE. In Table 7, σ_p is evaluated for SPE and OPE for $\gamma = 0.5$ with process noise consideration.

4 Conclusion

This paper presents a method for estimating thermal parameters, especially the convective heat transfer coefficient, based on thermography and DA. The EnKF is used to extract

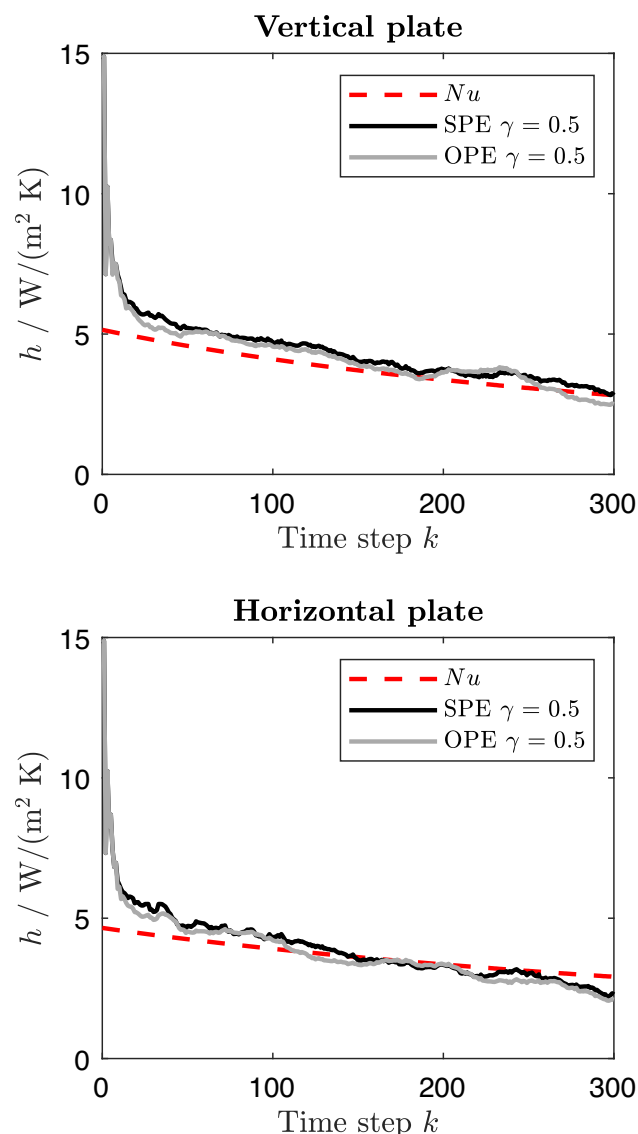


Fig. 9 Estimation values for h with real thermography measurement data - SPE and OPE with process noise consideration

Table 6 Pooled standard deviation for OPE without process noise consideration

Setup	Type	γ	$\sigma_p, [\sigma_p] = W/(m^2 K)$
VP	OPE	0.8	0.043
VP	OPE	0.5	0.035
VP	OPE	0.2	0.044
HP	OPE	0.8	0.063
HP	OPE	0.5	0.076
HP	OPE	0.2	0.220

Table 7 Pooled standard deviation for SPE and OPE with process noise consideration

Setup	Type	γ	$\sigma_p, [\sigma_p] = W/(m^2 K)$
VP	SPE	0.5	0.080
VP	OPE	0.5	0.090
HP	SPE	0.5	0.109
HP	OPE	0.5	0.256

parameters of interest from measured temperature data. Since convection processes need to take place for the estimation, the component is brought out of thermal equilibrium with its environment beforehand. When the process noise is not considered, a combination of SPE and PPE was found to give the best results. Especially at the beginning, the former implementation is used to compensate errors caused by the wrong parameter value on the spot. Once the parameter reaches its true value, the proposed method uses the latter implementation to focus the errors entirely on changing the parameter. Lowering the smoothing parameter can also lead to an improvement in the estimation result, but the probability of overshooting also increases. SPE and OPE deliver similar results taking the process noise into account.

Compared to classical parameter estimation methods the advantage of this method is its adaptability to the actual problem without the requirement of having full or exact knowledge about the underlying model and its parameters. As seen in Appendix A, different Nusselt correlations have to be applied for each geometric shape and its orientation. In the proposed method, it is sufficient to adapt only the geometric shape of the dynamic model to the real component. In this article, thermally thin components were studied. In case of $Bi \geq 1$, the computation domain of the 2D plate can be exchanged with a 3D shell to regain validity. Furthermore, much more complex problems can be handled, since a location-dependent temperature field can also be processed, or active thermography experiments by formulating the excitation in the boundary conditions. Another advantage is that the iterative computation is memory efficient, i.e. it can be used to estimate state and parameters at each time step without the need to store the complete data. This allows the application of the method for inline monitoring of long lasting processes.

Interesting topics, which subsequently arise from this work, relate to the measurement. As mentioned in the introduction, the EnKF can handle higher measurement noise compared to the noise encountered here, which enables the utilization of infrared cameras with uncooled detectors. In this case, it is interesting to identify the extent to which corrections for temperature drift are required. Furthermore, measurements from additional sensors could be processed in the EnKF in order to be able to handle more complex processes. For example, to improve the estimation accuracy in a homogenous high temperature environment, e.g.,

in an industrial autoclave, where ϵ is not close to one, a nonlinear correction function could be incorporated into the measurement equation $\mathbf{h}_k$. This function corrects the radiation measured by the camera using external thermocouples that measure the temperature of the background, e.g., the autoclave wall, and the air temperature.

Appendix A Nusselt Correlations

The Nusselt number

$$Nu = \frac{h L_{char}}{\lambda_F} \quad (19)$$

connects the Nusselt correlations to the investigated parameter h , in which L_{char} denotes the characteristic length and λ_F the thermal conductivity of the fluid. Depending on the type of convection, special quantities are needed for the Nusselt correlations. In the case of natural convection, these are the Grashof number

$$Gr = \frac{L_{char}^3 g \beta (T - T_\infty)}{\nu^2} \quad (20)$$

and Prandtl number

$$Pr = \frac{\nu \rho c_p}{\lambda} \quad (21)$$

as well as the product of both quantities, the Rayleigh number Ra . Here g is the gravitational acceleration, β the thermal volume expansion coefficient at a reference temperature and ν the kinematic viscosity of the fluid. For further information, reference is made to [3]. For the VP, the Nusselt correlation

$$Nu = \left\{ 0.825 + \frac{0.387 Ra^{1/6}}{[1 + (0.492/Pr)^{9/16}]^{8/27}} \right\}^2 \quad (22)$$

is valid for the laminar as well as for the turbulent flow ($0.1 < Ra < 10^{12}$). For the HP, a distinction is made between a hot and cold plate and between the upper and lower surface. For the upper surface of a hot plate and the lower surface of a cold plate, the Nusselt correlation

$$Nu = \frac{0.766 Ra^{1/5}}{[1 + (0.322/Pr)^{11/20}]^{4/11}} \quad (23)$$

is valid for the laminar flow ($Ra f_2(Pr) \lesssim 7 \cdot 10^4$) and the Nusselt correlation

$$Nu = \frac{0.15 Ra^{1/3}}{[1 + (0.322/Pr)^{11/20}]^{20/33}} \quad (24)$$

is valid for the turbulent flow ($Ra f_2(Pr) > 7 \cdot 10^4$). For the lower surface of a hot plate and the upper surface of a cold plate, the Nusselt correlation

$$Nu = \frac{0.6 Ra^{1/5}}{[1 + (0.492/Pr)^{9/16}]^{16/45}} \quad (25)$$

is valid ($10^3 < Ra f_1(Pr) < 10^{10}$).

Appendix B Reduced Measurement Sampling Rate

In the following, the scenario is considered when the measuring instrument does not fulfill the required sampling rate f from Section 3. Specifically, the case of half the sampling rate, i.e., $f = 0.5$ Hz, is examined on the measurement data

from the simulation study. Figure 10 shows the estimation results for the convective heat transfer coefficient h obtained by OPE with different values of the smoothing parameter γ , with a similar behavior to Fig. 4.

Furthermore, the process noise is taken into account in Fig. 11 analogous to Fig. 5. Again, the same behavior occurs. Independent of the chosen sampling rate, the workflow requires a certain number of steps k to reach the actual value of the searched parameter from a generally assumed value, e.g., Table 3. Consequently, a sufficient sampling rate is necessary in cases, in which the searched parameter changes significantly or generally assumed values are specified as initial values, as in this article.

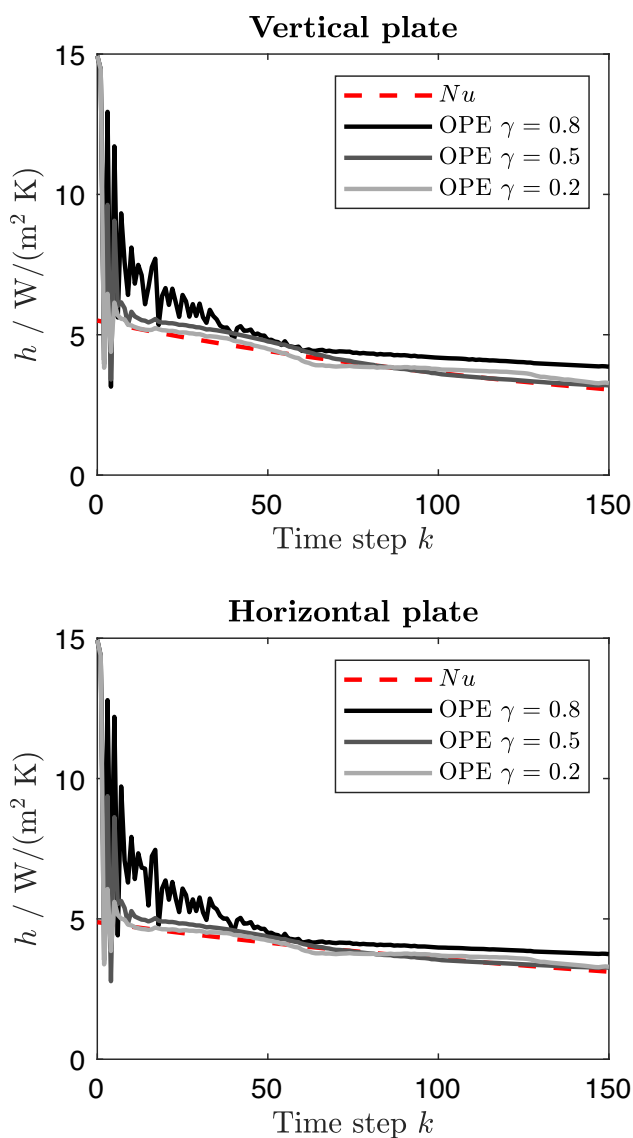


Fig. 10 Estimation values for h for a reduced measurement sampling rate within the simulation study - Variation of the smoothing parameter γ with OPE

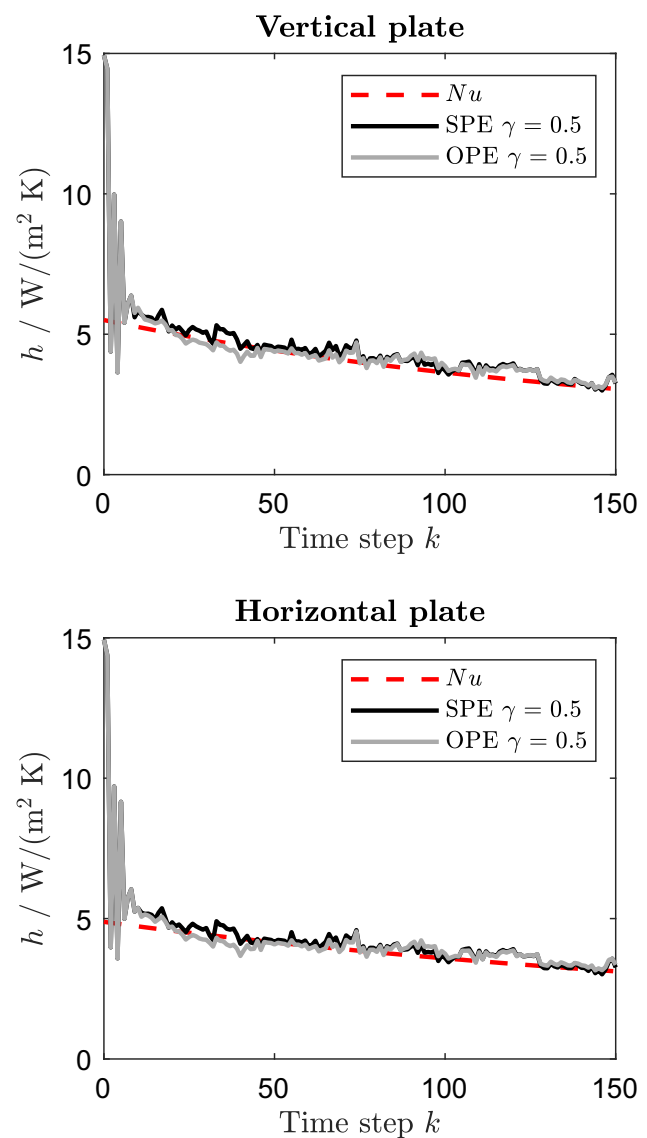


Fig. 11 Estimation values for h for a reduced measurement sampling rate within the simulation study - Comparison of SPE and OPE with process noise consideration

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Materials availability Not applicable.

Code availability Not applicable.

Declarations

Competing Interests The authors declare no competing interests.

Ethics approval and consent to participate Not applicable.

Consent for Publication Not applicable.

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